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1. $y = -(x+3)(x-1)$

(i) zero product rule :

$x + 3 = 0$ $\qquad$ $x - 1 = 0$

$x = -3$ $\qquad$ $x = 1$

Coordinater : $(-3, 0)$ $(1, 0)$

(ii) C is y-intercept

$x = 0$

$y = -(x+3)(x-1)$

$= -(0+3)(0-1)$

$= -(3)(-1)$

$= 3$

cordinater : $(0, 3)$

A $(-3, 0)$

B $(1, 0)$

C $(0, 3)$

(iii) $x = -1$ is the line of symetry.

Line of symetry is peak or lowest.

substitute $x = -1$ into $y = -(x+3)(x-1)$

$= -(-1+3)(-1-1)$

$= -(2)(-2)$

$= 4$

Maximum point = $(-1, 4)$

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2. $y = (x-4)(x+2)$ cuts in point A, B, C

- A and B on X
- C on Y

(i) Zero product principle

$x - 4 = 0$ $\quad$ $x + 2 = 0$

$x = 4$ $\quad$ $x = -2$

Coordinates : $(4, 0)(-2, 0)$

! C is y-intercept

$x = 0$ $\quad$ $y = (x-4)(x+2)$

$= (0-4)(0+2)$

$= (-4)(2)$

$= -8$

Coordinates : $(0, -8)$

A $(4, 0)$

B $(-2, 0)$

C $(0, -8)$

(ii) P $(-3, k)$ lier on curve

$\downarrow \quad \downarrow$

$x \quad y$

$x = -3$

$k = (x-4)(x+2)$

$k = (-3-4)(-3+2)$

$k = (-7)(-1)$

$k = 7$

P $(-3, 7)$

(iii) Explain why the point Q doer not lies on the curve.

Q $(3.5, -2.5)$

Q $(\frac{7}{2}, -\frac{5}{2})$

$x = .$ $\quad$ $y = (x-4)(x+2)$

$= (3.5-4)(3.5+2)$

$= (-0.5)(5.5)$

$= -2.75$

Because $y = -2.75 \neq -2.5$

Q does not lier on curve.

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3. The variables x and y are connected by the equation $y = x^2 - 3x - 2$.
Some corresponding values of x and y are given in the table below.

x	−2	−1	0	1	2	3	4
y	p	2	−2	−4	−4	−2	2

p = 8

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-2 \leqslant x \leqslant 4$.

(c) Use your graph in part (b) to find

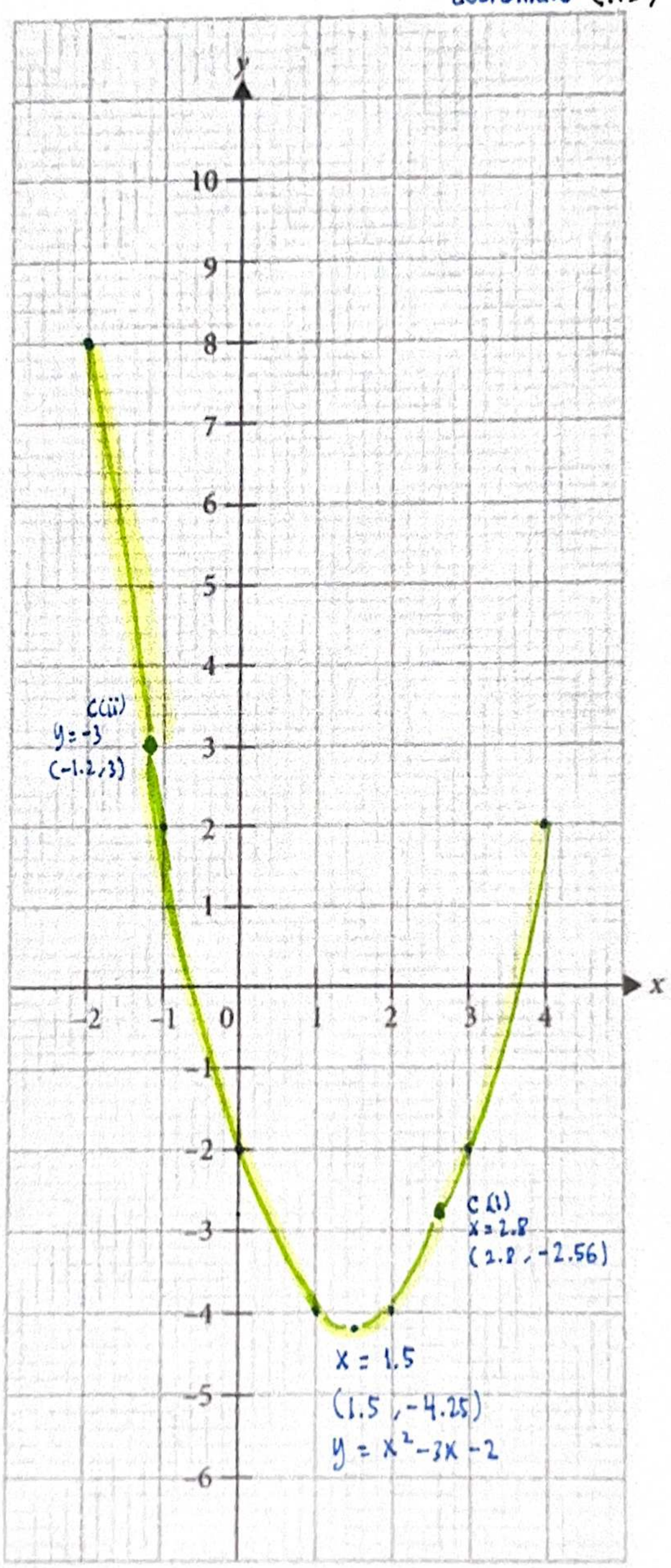
(i) the value of y when $x = 2.8$, y = -2.56

(ii) the values of x when $y = -3$, y = -2

(iii) the minimum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph. x = 1.5

coordinate (1.5, -4.25)



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3. $y = x^2 - 3x - 2$

a. Find value of p if $x = -2$

x	-2	
x^2	$(-2)(-2) = 4$	$p = 8$
$3x$	$3(-2) = -6$	
$x^2 - 3x$	$4 - (-6) = 10$	
$x^2 - 3x - 2$	$10 - 2 = 8$	
y	8	

c. (iii) minimum point

(a) Find $\boxed{x = -\frac{b}{2a}}$

$a = 1$ $\quad x = \frac{-(-3)}{2(1)}$

$b = -3$

$c = -2$ $\quad x = \frac{3}{2} = 1.5$

(b) subs x into y equation.

Subs $x = 1.5$ into $y = x^2 - 3x - 2$

$= (1.5)^2 - 3(1.5) - 2$

$= (1.5)(1.5) - 3(1.5) - 2$

$= (2.25) - (4.5) - 2$

$= -4.35$

Minimum coordinates = $(1.5, -4.35)$

4. The variables x and y are connected by the equation $y = -x^2 + 2x + 1$.
Some corresponding values of x and y are given in the table below.

x	−1	−0.5	0	0.5	1	2	3
y	−2	p	1	1.75	2	1	−2

$p = -0.25$

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-1 \leq x \leq 3$.

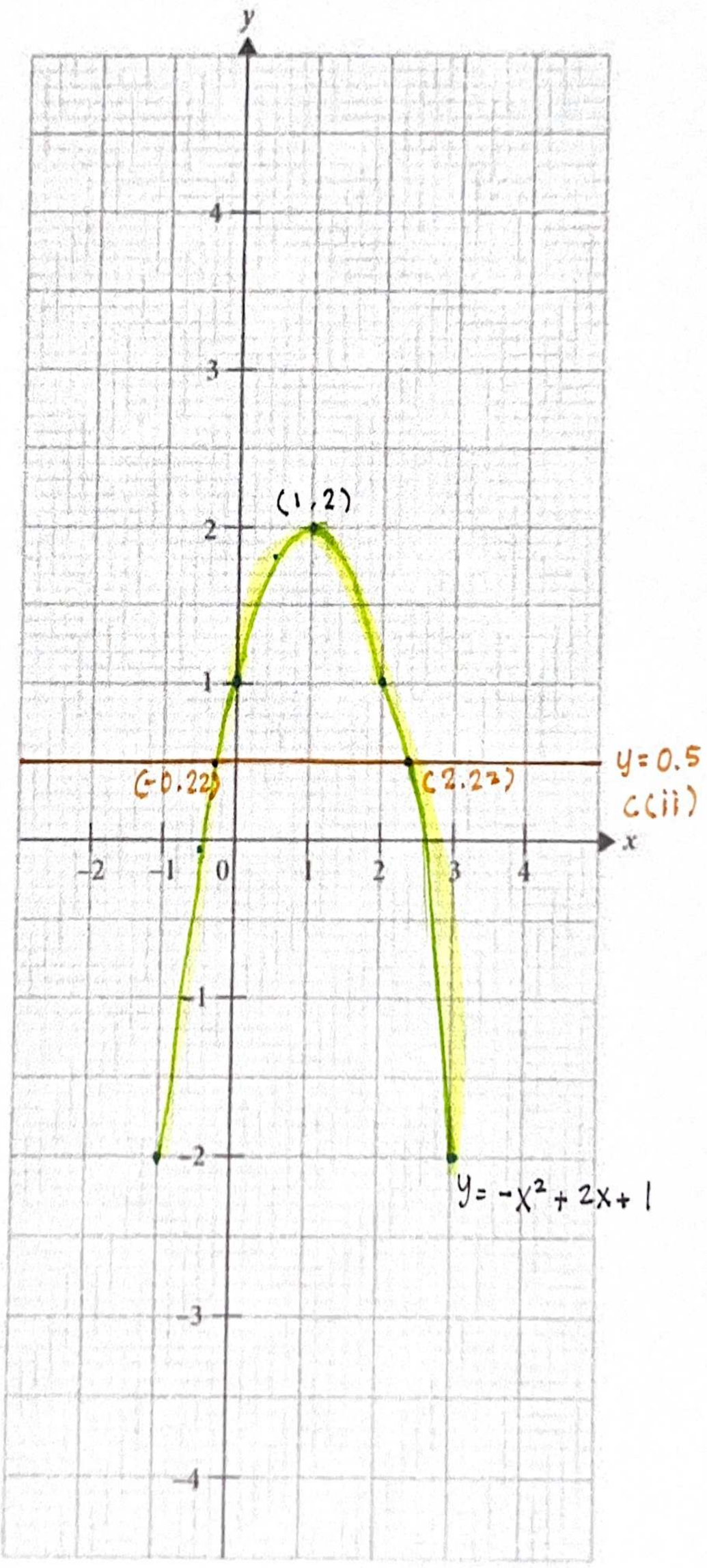
(c) Use your graph in part (b) to find

(i) the value of y when $x = -0.7$,

(ii) the values of x when $y = 0.5$,

(iii) the maximum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph.



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4. $y = -x^2 + 2x + 1$

a. Find the value of p if $x = -0.5$

x	-0.5	p = -0.25
$-x^2$	$-(-0.5)(-0.5) = -0.25$	
$2x$	$2(-0.5) = -1$	
$-x^2 + 2x$	$-0.25 + (-1) = -1.25$	
$-x^2 + 2x + 1$	$-1.25 + 1 = -0.25$	
y	-0.25	

c. (iii) The maximum value

$a = -1$

$b = 2$

$c = 1$

$$x = \frac{-b}{2a}$$

$$x = \frac{-(2)}{2(-1)}$$

$$x = \frac{-2}{-2} = 1$$

subs $x = 1$

$$y = -x^2 + 2x + 1$$
$$= -(1)^2 + 2(1) + 1$$
$$= -1 + 2 + 1$$
$$= 2$$

Maximum value coordinate $(1, 2)$

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1 dp

4. c (i) Use your graph to find the value of y when $x = -0.7$

$y = -x^2 + 2x + 1$

$= -(-0.7)^2 + 2(-0.7) + 1$

$= -0.9$

coordinater : (-0.7 , -0.9)

c (ii) Use your graph to find the value of x when $y = 0.5$

$y = -x^2 + 2x + 1$

$0.5 = -x^2 + 2x + 1$

$0 = -x^2 + 2x + 1 - 0.5$

$0 = -x^2 + 2x + 0.5$ x -1

$x^2 - 2x - 0.5 = 0$

$a = 1$

$b = -2$

$c = -0.5$

$$x_{1.2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_{1.2} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-0.5)}}{2(1)}$$

$x_1 = 2.2$ $x_2 -0.2$

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5. (a) Complete the table.

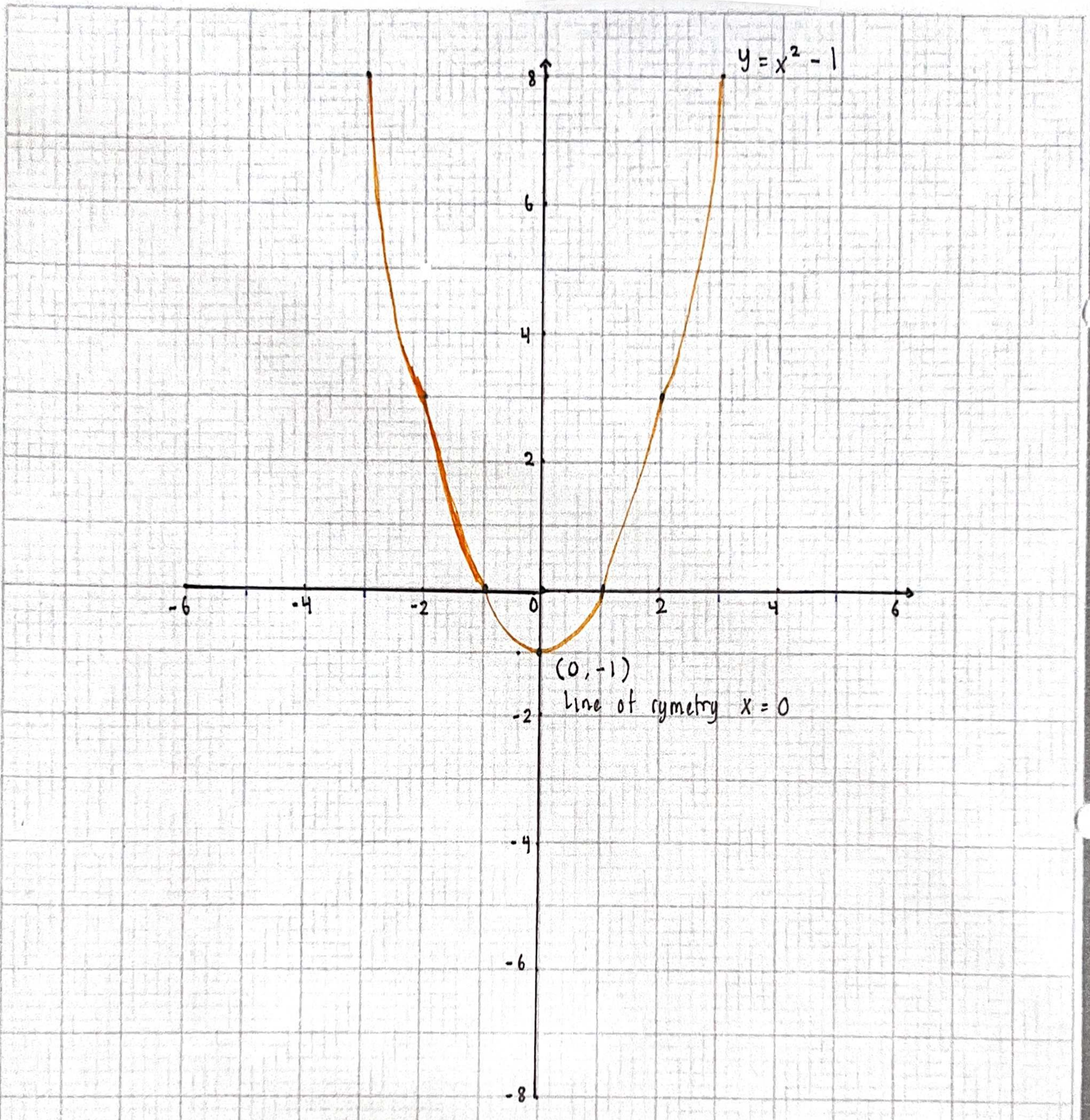
x	−3	−2	−1	0	1	2	3
$y = x^2 - 1$	8	3	0	-1	0	3	8

(b) Using a scale of 2 cm to represent 1 unit on both axes, draw the graph of $y = x^2 - 1$ for $-3 \leq x \leq 3$.

(c) Use your graph in part (b) to find

(i) the coordinates of the minimum point of the graph.

(ii) the equation of the line of symmetry.



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6. (a) Complete the table.

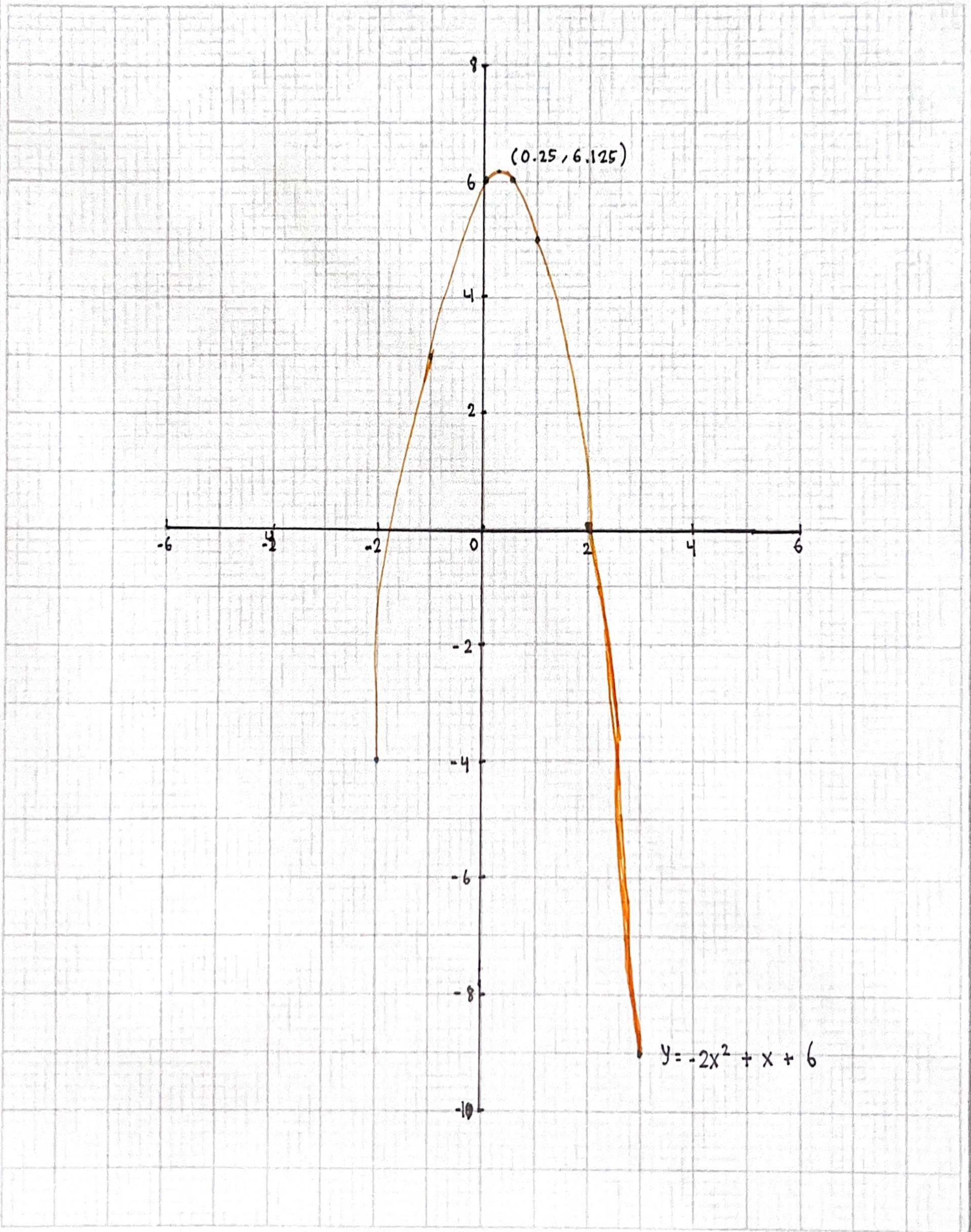
x	−2	−1	0	0.5	1	2	3
$y = -2x^2 + x + 6$	-4	3	6	6	5	0	9

(b) Using a scale of 2 cm to represent 1 unit on the x-axis and 1 cm to represent 1 unit on the y-axis, draw the graph of $y = -2x^2 + x + 6$ for $-2 \leq x \leq 3$.

(c) Use your graph in part (a) to find

(i) the coordinates of the turning point of the graph, stating whether it is a maximum point or a minimum point,

(ii) the equation of the line of symmetry of the graph.



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6. c (i) minimum / maximum point invertigation

$y = -2x^2 + x + 6$

$a = -2$

$b = 1$

$c = 6$

$X = \frac{-b}{2a}$

$X = \frac{-(1)}{2(-2)}$

$= \frac{-1}{-4} = \frac{1}{4} = 0.25$

$X =$ $y = -2x^2 + x + 6$

$= -2(\frac{1}{4})^2 + (\frac{1}{4}) + 6$

$= -2(\frac{1}{4})(\frac{1}{4}) + (\frac{1}{4}) + 6$

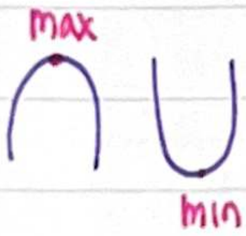
$= -(\frac{1}{8}) + (\frac{1}{4}) + 6$

$= 6.125$

Coordinates (0.25 , 6.125)

Value of x is range $-2 \leq x \leq 3$

y is $y = -2x^2 + x + 6$



Because = a = -2 < 0, the graph opens downward / upward

therefore it is maximum / minimum

C (ii) Peak / Maximum point (0 , 6)

The equation of line of symetry (x)

$X = 0$

7. A gardener has 70 metres of fencing to make a rectangular enclosure.
The length of the enclosure is x metres.

(a) Show that the area, A m^2, of the enclosure is given by $A = x(35 - x)$.

(b) This is a table of values for $A = x(35 - x)$.

x	0	5	10	20	30	35
A	0	150	250	p	150	0

(i) Find the value of p.

(ii) Draw the graph of $A = x(35 - x)$.
Use a scale of 2 cm to 5 m on the horizontal x-axis.
Use a scale of 1 cm to 20 m^2 on the vertical A-axis.

(c) Use your graph to find

(i) the value of x when A is a maximum,

(ii) the maximum value of A.



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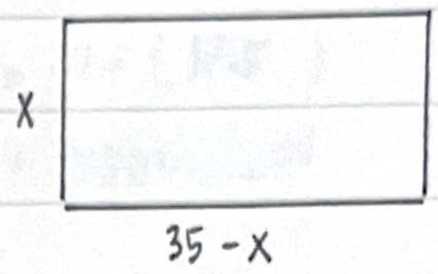
7. A gardener has 70 metres of fencing.

(a) Show that the area, A m^2, of the enclosure is given by $A = x(35 - x)$

Fence = Parameter

$70 = 2(L + w)$

$70 = 2L + 2W$



(i) simplify the parameter

$70 = 2L + 2W$ ÷ 2

$35 = L + W$

(ii) Find W

$L + W = 35$

$W = 35 - L$

(iii) Put into Area

$A = L \cdot W$ $\quad L = X$

$A = L(35 - L)$

shows that the Area is $A = x(35 - x)$

$A = x(35 - x)$

(b) This table is for the Area

$A = x(35 - x)$

$P = 300$

X	20
35 - X	35 - 20 = 15
X (35 - X)	20 (35 - 20) = 20 (15) = 300
A or P	300

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7. C. The value of x when A maximum.

(i) Find x

$$X = -\frac{b}{2a}$$

$$X = -\frac{(35)}{2(-1)}$$

$$X = 17.5$$

(ii) Put x into y.

$y = A$

$A = x(35 - x)$

$A = 17.5(35 - 17.5)$

$= 17.5(17.5)$

$= 306.25$ (2 dp)

maximum A = 306.25 (2dp)

value of X = 17.50 (2dp)

Where's my a and b ?

To find a and b

expand the equation of A.

$A = x(35 - x)$

$A = 35x - x^2$

reorder

$A = -x^2 + 35x$

$a = -1$ $b = 35$ $c = 0$

8. The height, h metres, of a ball projected vertically upwards from a ball machine can be modelled by the equation $h = -4t^2 + 12t$, where t is the time in seconds after it leaves the machine.

(a) (i) Solve the equation $-4t^2 + 12t = 0$.

(ii) Write down the duration of flight of the ball.

(b) Draw the graph of $h = -4t^2 + 12t$ during the duration of flight.

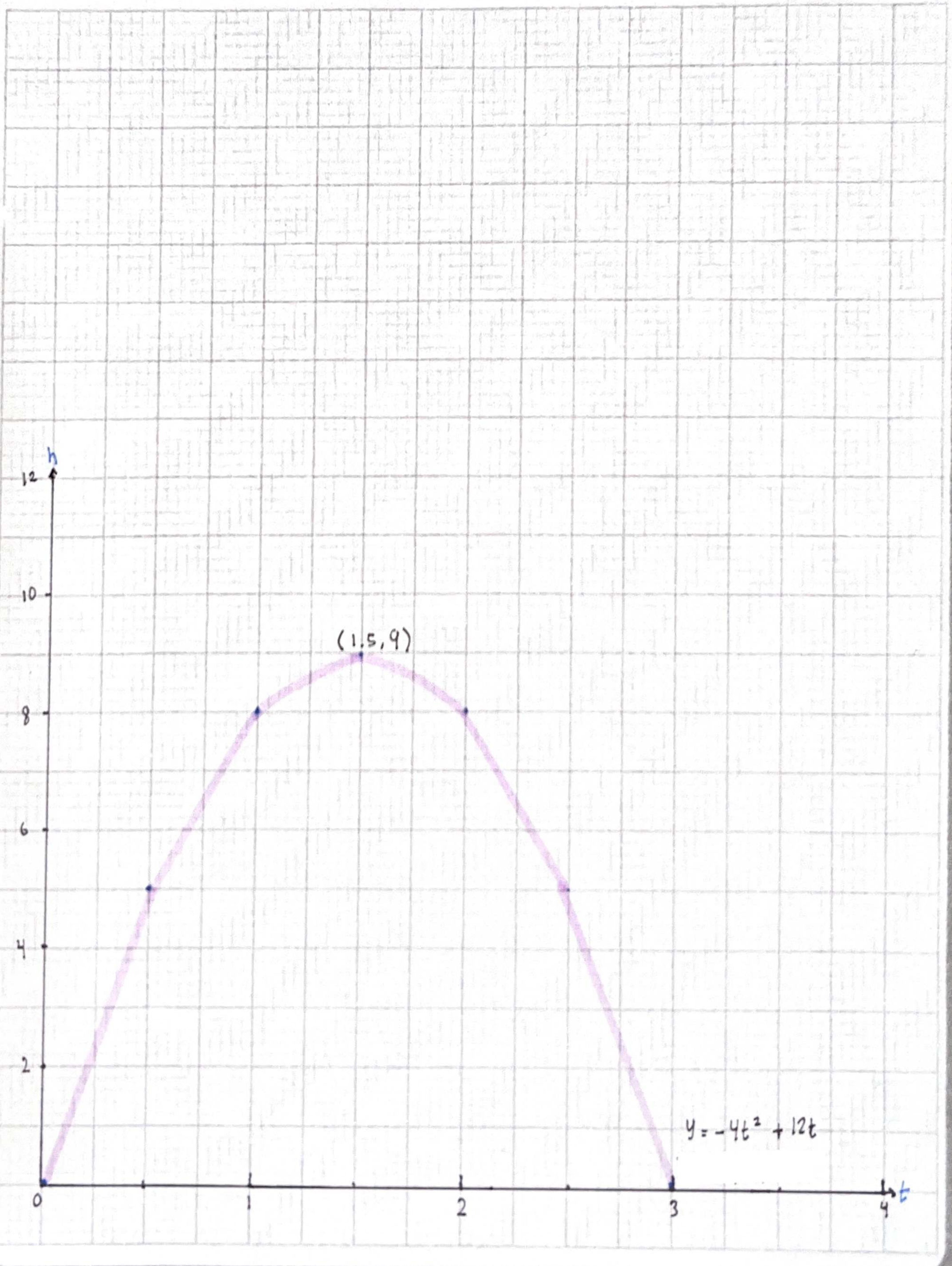
Use a scale of 4 cm to 1 s on the horizontal t-axis.

Use a scale of 2 cm to 1 m on the vertical h-axis.

(c) Use your graph to find

(i) the maximum height of the ball above the ground,

(ii) the value(s) of t when the ball is 6 m above the ground.



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8. The height, h metres, of a ball projected vertically upwards from a ball machine can be modelled by the equation $h = -4t^2 + 12t$, where t is the time in seconds after it leaves the machine.

(a) (i) Solve the equation $-4t^2 + 12t = 0$.

(ii) Write down the duration of flight of the ball.

(b) Draw the graph of $h = -4t^2 + 12t$ during the duration of flight.

Use a scale of 4 cm to 1 s on the horizontal t-axis.

Use a scale of 2 cm to 1 m on the vertical h-axis.

(c) Use your graph to find

(i) the maximum height of the ball above the ground,

(ii) the value(s) of t when the ball is 6 m above the ground.

8 a. Solve the equation

means find the factors

$-4t^2 + 12t = 0$

Note:

1. Try to simplify
2. Factors
   - using identity

     $(a+b)^2$ or $(a-b)^2$ or a^2-b^2 or zero product principle
   - fill out commons factors

     $ab + ac = a(b+c)$ | $-ab + ac = -a(b-c)$

     $ab - ac = a(b-c)$ | $-ab - ac = -a(b+c)$
   - zero product principle $-3(x-2)(x+3)$

     $x-2=0$ $x+3=0$

$\frac{-4t^2 + 12t = 0}{} \div 4$

$-t^2 + 3t = 0$ (id) $-ab + ac = -a(b-c)$

$-t(t-3) = 0$

$t = 0$ $t \Rightarrow (t-3) = 0$

$t = 3$

Factor

$t = 0$ or 3

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8.

a (ii) Duration of flight

To know the range $y \leq x \leq y$

factors of $-4t^2 + 12t = 0$

$t = 0$ or 3

t	min: 0	0.5	1	1.5	2	max: 3
h	0	5	8	9	8	0

b. Remember :

t-axis scale of 4 cm for every 1 s

h-axis scale of 2 cm for every 1 s

c. Maximum height.

$-4t^2 + 12t = 0$

$a = -4$

$b = 12$

$c =$

$$x = \frac{-b}{2a}$$

$$x = \frac{-(12)}{2(-4)}$$

$x = 1.5$

$h = -4(1.5)^2 + 12(1.5)$

$= -4(1.5)(1.5) + 12(1.5)$

$= -9 + 18$

$= 9$

Maximum point (x, h) : $(1.5, 9)$

maximum height : 9 m

e (ii) Value of (s) of t when ball is 6 m above the ground.

$-4t^2 + 12t = h$

$-4t^2 + 12t =$

$-4t^2 + 12t - 6 = 0$

simplify (same factors)

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8 c (ii) $-4t^2 + 12t - 6 = 0$ ÷ 2

$-2t^2 + 6t - 3 = 0$

hard factor

Hard factors use:

$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$

↓

Change var

$t_{1,2} = \frac{-(6) \pm \sqrt{(6)^2 - 4(-2)(-3)}}{2(-2)}$

$= \frac{3 \pm \sqrt{3}}{2}$

The value of t

t = 2.4 or 0.6

$t_1 = 2.4$ $t_2 = 0.6$

9. The shape of an underground passageway can be modelled by the equation $y = -0.02x^2 + 2x - 68$, where x is the horizontal distance in metres from one entrance of the passageway and y is the depth below ground level in metres.

The passageway extends 100 m horizontally.

(a) Draw the graph of $y = -0.02x^2 + 2x - 68$ for $0 \leq x \leq 100$.

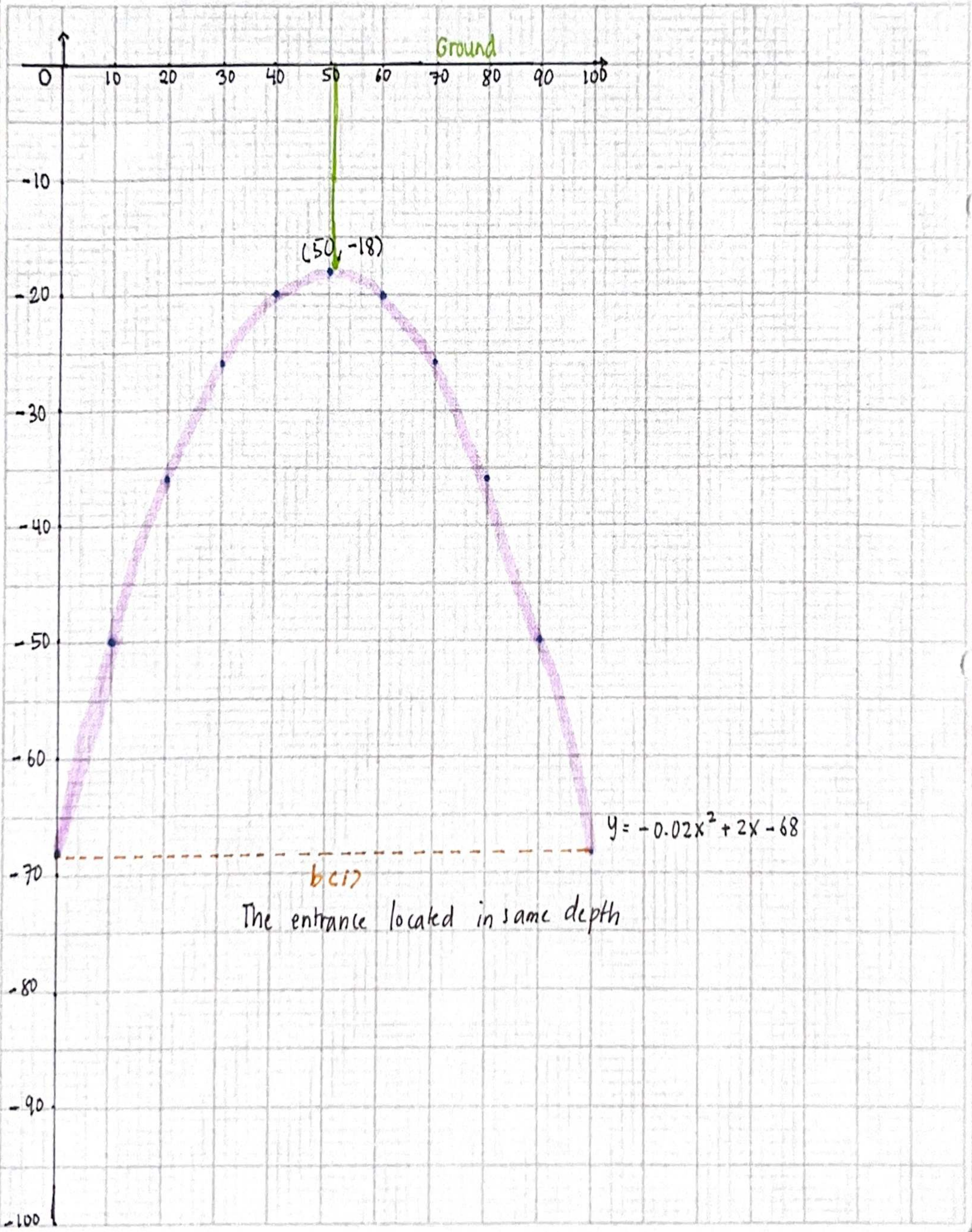
Use a scale of 1 cm to 10 m on the horizontal x-axis.

Use a scale of 2 cm to 10 m on the vertical y-axis.

(b) Use your graph to find

(i) whether both entrances of the passageway are located at the same depth,

(ii) the minimum height difference between the passageway and the ground.



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9. The shape of an underground passageway can be modelled by the equation $y = -0.02x^2 + 2x - 68$, where x is the horizontal distance in metres from one entrance of the passageway and y is the depth below ground level in metres.

The passageway extends 100 m horizontally.

(a) Draw the graph of $y = -0.02x^2 + 2x - 68$ for $0 \leq x \leq 100$.
Use a scale of 1 cm to 10 m on the horizontal x-axis.
Use a scale of 2 cm to 10 m on the vertical y-axis.

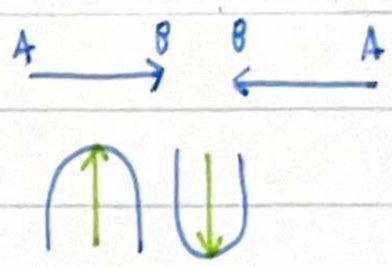
(b) Use your graph to find
(i) whether both entrances of the passageway are located at the same depth.
(ii) the minimum height difference between the passageway and the ground.

9.

$y = -0.02x^2 + 2x - 68$

x = horizontal distance

y = the depth



the passageway extend 100 m horizontally.

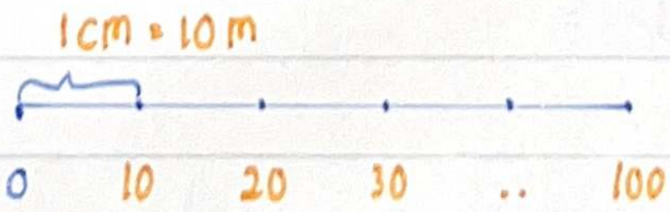
- no added start point, thinking $x = 0$ $0 \leq x \leq 100$

X	0	10	20	30	40	50	60	70	80
y	-68	-50	-36	-26	-20	-18	-20	-26	-36

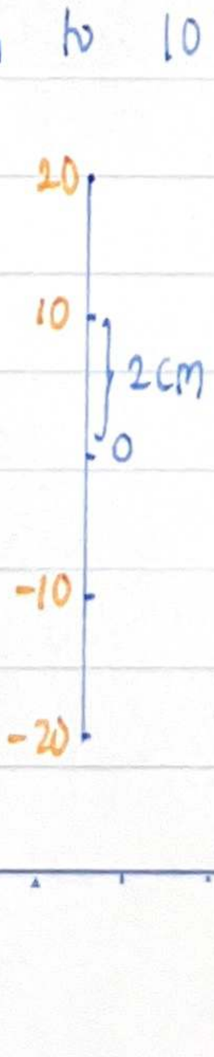
X	90	100
y	-50	-68

(using scale)

1 cm to 10 m on horizontal axis (x)



2 cm to 10 m on vertical y-axis (y)



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Use your graph !!

b. Use your graph to find :

(i) Whether both entrance located in same depth

look : The passageway extends 100 m horizontally

$0 \leq x \leq 100$

means there $x = 0$ } 2 as entrance.
$x = 100$

based on table

$x = 0 \quad y = -68$

$x = 100 \quad y = -68$

(ii) The minimum height difference between passageway and the ground. The minimum and maximum.

a. Find x

$y = -0.02x^2 + 2x - 68$

$x = \frac{-b}{2a} \quad x = -\frac{(2)}{2(-0.02)} = \frac{-2}{-0.04} = 50$

$a = -0.02$

$b = 2$

$c = -68$

b. Find the point of symetry

$x = 50$

$y = -0.02x^2 + 2x - 68$

$= -0.02(50)^2 + 2(50) - 68$

$= -0.02(50)(50) + 2(50) - 68$

$= -18$

Because y present as negative number,

the passageway is 18 m (below) / upper ground.